Coding with AI

Context, workflow, and using AI in science

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The plan

1. **Tokens and context** — the window, and the handoff that curates it
2. **From chat to agentic coding** — the agent finds its own context
3. **How I code with an agent** — foundation · spec · slices · evidence
4. **Skills as maintained assets** — package the disciplines you repeat
5. **Ownership and validation** — the judgement that stays human

What is a token?

The smallest bit of information the model works with.

A node connected to the other tokens in the window.

Its working meaning comes from those connections.

How many r's in strawberry?

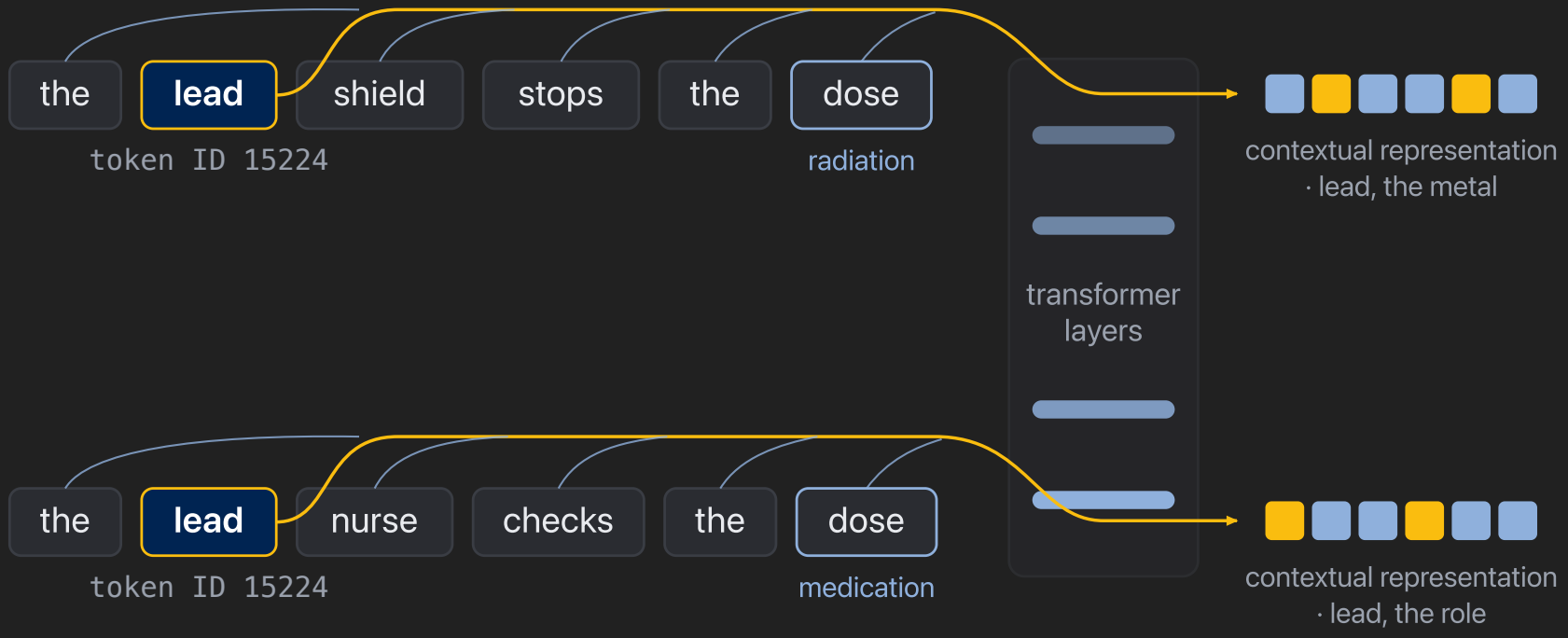
`strawberry`

A direct count asks the model to work below its own units: it sees chunks, not letters.

`s t` **`r`** `a w b e` **`r r`** `y`

> When a model struggles: **change the representation**, expose intermediate steps, or give it a tool.

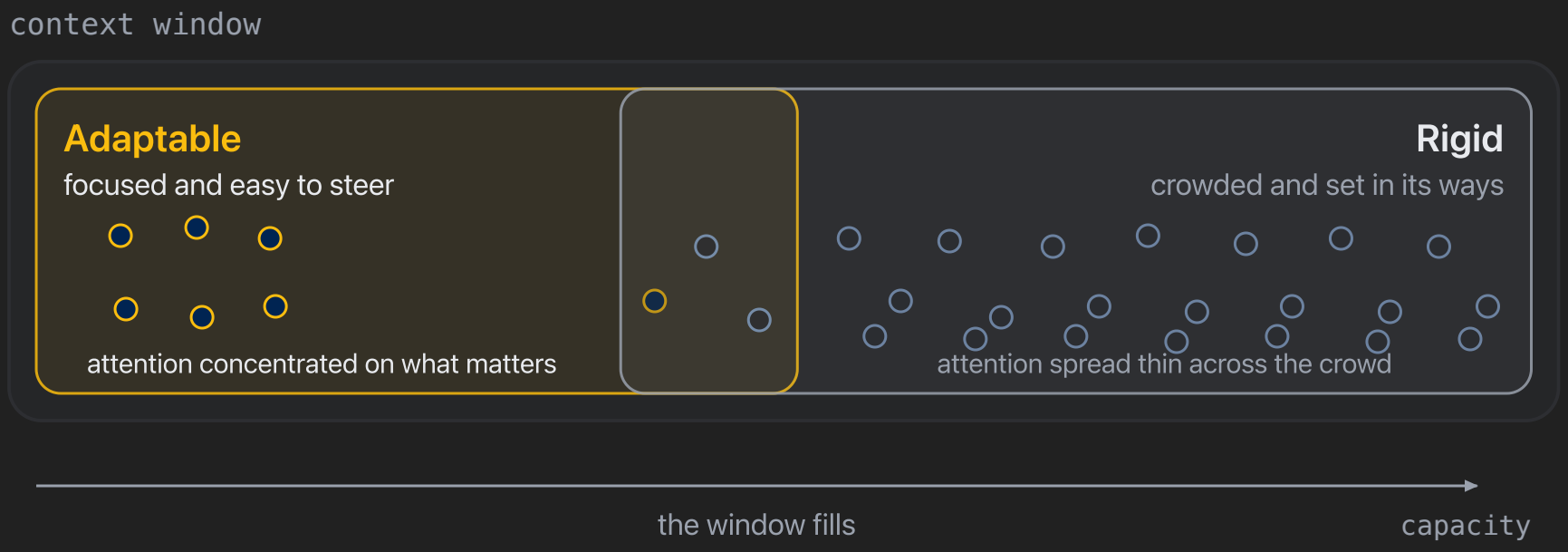
Same token ID, different meaning



Same ID in. Different working meaning out.

> The token ID is fixed. The representation carried at its position becomes **contextual**.

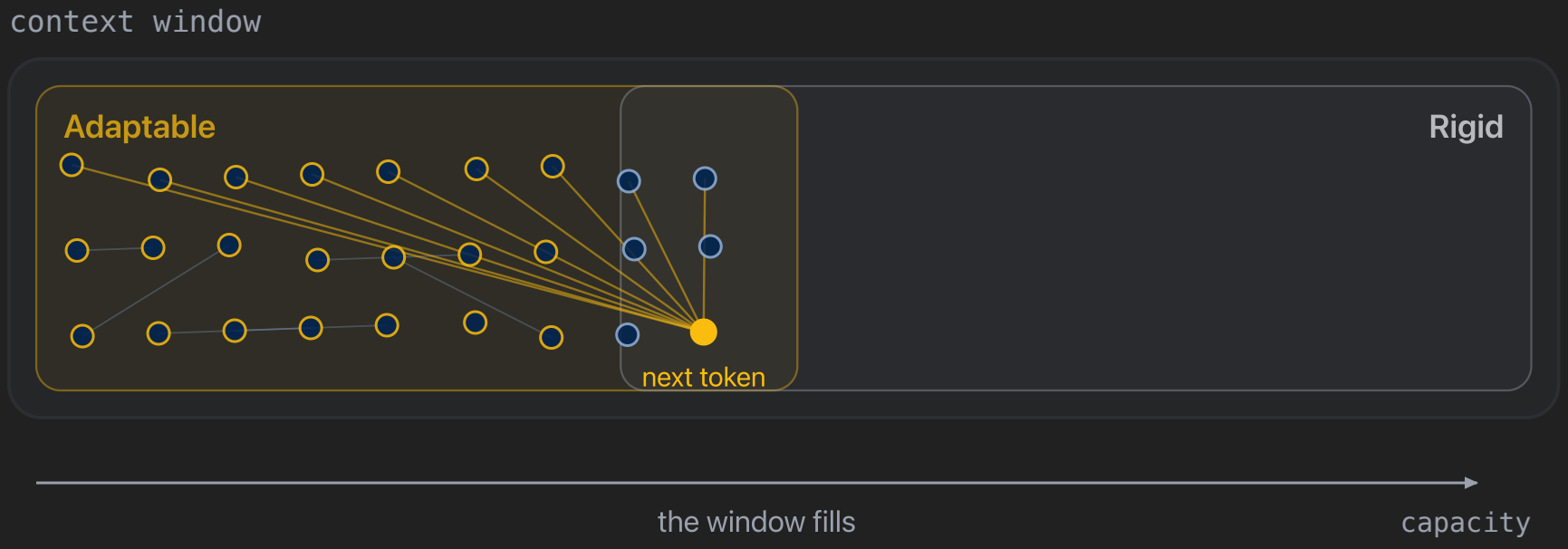
Keep context small and relevant



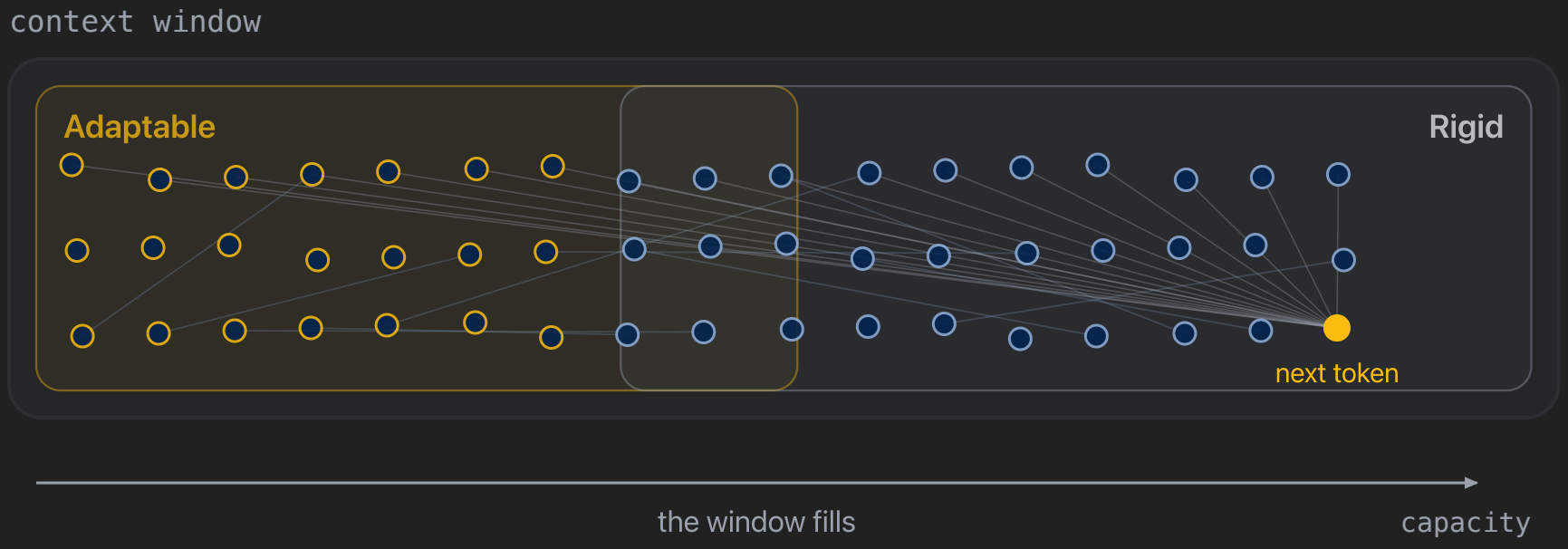
A million-token window is capacity, not a target.

A fresh chat: small and targeted

The chat grows



Near capacity: attention spread thin



> A model's best work happens when the context is **small and targeted**.

Do not be afraid to start a new chat

New chat

A fresh, focused window. Steering is easiest before the crowd moves in.

/btw side questions

Answered in an overlay: it sees the whole chat, but the answer never enters the window.

Carrying the thread into the new chat is a handoff.

A workspace the agent can navigate

Your workspace: a Project in the web app, or a codebase on disk.

One huge file

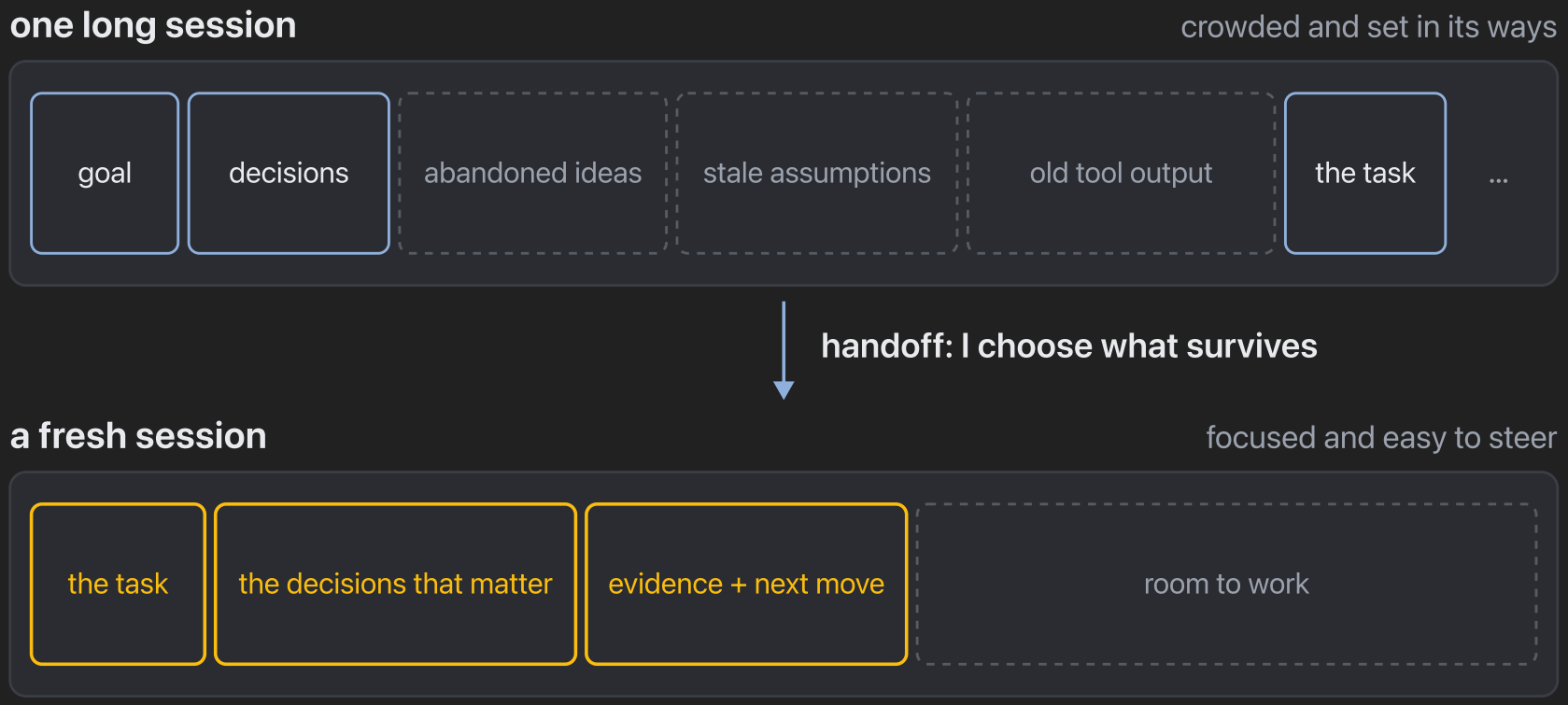
The whole textbook in a single document. No window can hold it.

Navigable chunks

A file per chapter section. The agent parses the structure, then loads only what the task needs.

Structure the workspace so any one task fits a small window.

Handoff: compaction with intention



One bounded task, only the context needed to complete it.

Automatic compaction vs handoff

automatic compaction

fires when the window is already full
a summary nobody curated can silently drop a locked decision
what was lost is invisible

handoff

happens when I choose
carries what the next task needs
decisions survive because I put them there
a file I can read and audit

> Both compress the conversation. Only one is **intentional**.

One handoff, in practice: a diverged pyKMC bug

- A pyKMC bug was already fixed, but on a heavily refactored branch
- The fix could not be transplanted into `develop`: different seams

```
carried:  the bug, narrowly stated
carried:  produce a failing test first
carried:  clues from the branch fix
carried:  the constraint, a small native change
```

> Carry the problem, evidence, and clues across the boundary, **not an obsolete solution**.

What a handoff carries depends on where it is going

Direction	Might carry
Continue the same work	current state · settled decisions · useful evidence · next move
Break in a new direction	relevant premise · divergence point · local boundary · open question
Execute one piece of a larger plan	pointer to the plan · scope of the piece · starting state · verification target
Separate worktree	branch pointer · integration boundary · relevant checks

> **Examples, not required boxes.** I choose what the next context needs.

Parallel work: handoffs + worktrees

```
same task:  stale context → curated handoff → fresh chat, same code state
one planned piece:  written plan → bounded handoff → fresh chat → verified result
parallel:  independent handoffs → separate worktrees → merge + integration checks
```

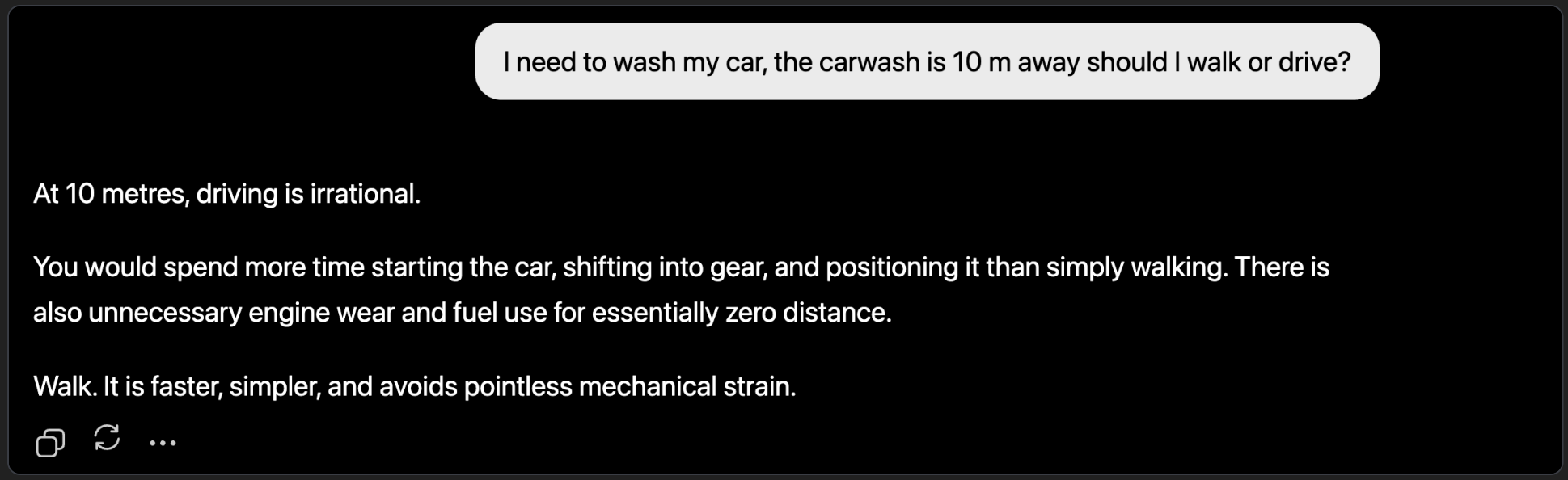
> A handoff scopes the **context**; a worktree isolates the **code state**. Neither makes integration automatic.

Vocabulary: tokens and context

Term	In this talk
token	The smallest bit of information the model works with. A node whose meaning comes from its connections.
token ID	The fixed integer produced by tokenization. It is not what changes.
contextual representation	The vector carried at a position after the layers mix in context. Two identical IDs can diverge.
context window	The model's finite working set. Select what should shape the next answer; do not fill it.
adaptable / rigid	My labels for session quality: focused and steerable, versus crowded and set in its ways.
/btw side question	Answered beside the chat: it sees the conversation, but never enters the window.
handoff	Compaction with intention: one bounded task, only the context needed to complete it.
automatic compaction	The safety net when the window fills: a summary nobody curated. Do not depend on it.
worktree	An isolated checkout. Separates code state, not conversational context.

“I need to wash my car, the carwash is 10 m away should I walk or drive?”

Walk or drive?



ChatGPT 5.3 Instant · 2026-07-13 · one observed response

The car has to arrive at the wash.

Walk… unless

I need to wash my car, the carwash is 10 m away should I walk or drive?

If the car wash is literally **10 meters away**, walking makes much more sense.

At that distance:

- Walking takes only a few seconds.
- Driving would likely take longer than the walk once you account for getting in, starting the car, and maneuvering it.

The one exception is if you mean you need to **bring the car to the car wash** because it's a drive-through or automated wash. In that case, you would, of course, drive the car the 10 meters into the wash.

So:

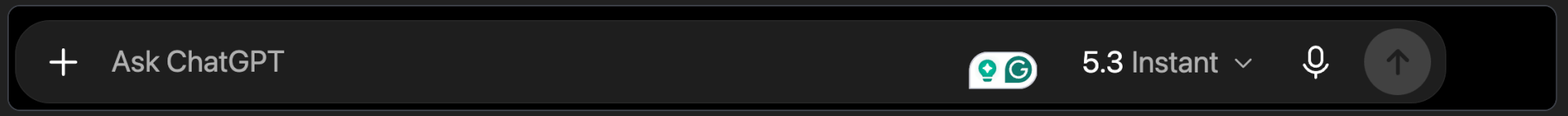
- If you're just going over there yourself: **walk**.
- If you're taking the car into the wash: **drive the car the 10 meters into the wash**.

ChatGPT 5.5 Instant · 2026-07-13 · one observed response

Same prompt, different answers

The same 10 m question, other models and modes (July 2026):

Response	What it did
ChatGPT 5.3 Instant	Walk. The purpose never surfaces. (shown live)
ChatGPT 5.4 Pro	Drive: the car must reach the wash.
ChatGPT 5.5 Instant	Walk first, then the exception: the car has to be brought in. (shown above)
Claude Haiku 4.5	Walk.
Claude Sonnet 5 Low	Walk.
Claude Opus Low	Drive.



> One correct answer to a familiar trap is **not evidence** the model will recover the hidden purpose of a novel request.

Inside the codebase

Web chat

Works from the context I give it. It sees the fragments I paste or upload, and nothing else.

Agentic coding

Works inside the codebase. It goes looking for the files and relationships the task needs, within the access it has been granted.

> An LLM working in my codebase, sussing out where things are for itself. CLI or app: **the interface is not what makes it agentic.**

I was the agent

- Two courses, before any of these tools had agency
- One total energy, buried in a huge Quantum ESPRESSO output
- A grad-course analysis whose output file swamped the chat
- Both worked only once I supplied the procedure:

```
grep the lines that matter → write them to a small file → analyse that → report back
```

I was doing a subagent's job by hand.

Vocabulary: agentic coding

Term	In this talk
agentic coding	An LLM working inside the codebase, having sub-workers find relevant context for itself.
situated access	The agent inspects the repository, bounded by the access it was granted.
inspect · act · observe · adapt	How a coding agent behaves once it can see the work. A description, not the boundary.
web chat	Works only from what you paste in. Capable, but context-blind.

Skill: build-design-foundation

FOUNDATION.md: a context start

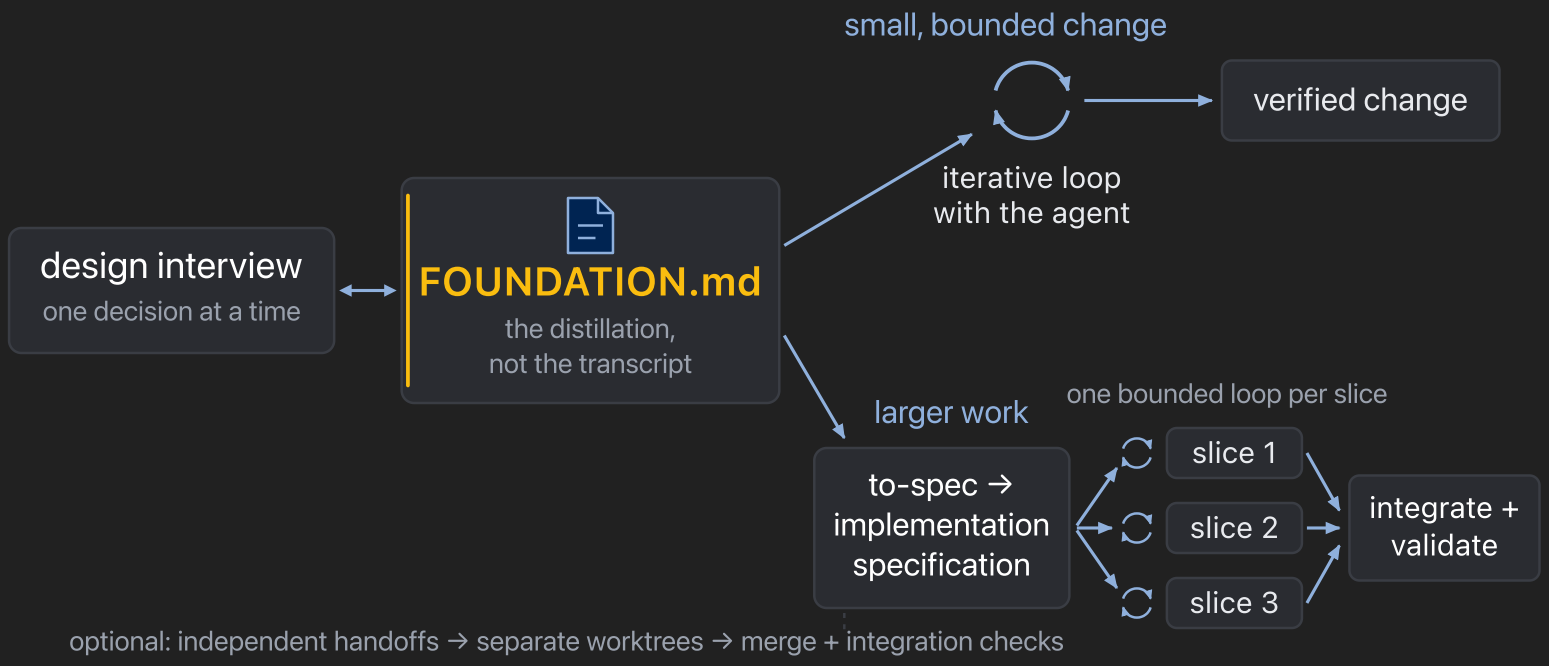
- The distillation, not the transcript
- Goal · glossary · locked decisions and why · boundaries · evidence · definition of done
- A new session loads the agreed design directly

> Do not depend on automatic compaction for durable design.
> **Commit the north star to a file.**

This presentation was built exactly this way.

Skill: to-spec

Scale the process to the work



> Every step hands the next agent a **smaller, sharper context**.

Ralph: the blunt end of the loop

```
while ! ./slices-done.sh; do
  claude -p "Read SPEC.md and PROGRESS.md.
             Take the next unfinished slice.
             Implement it, run its checks,
             update PROGRESS.md."
done
```

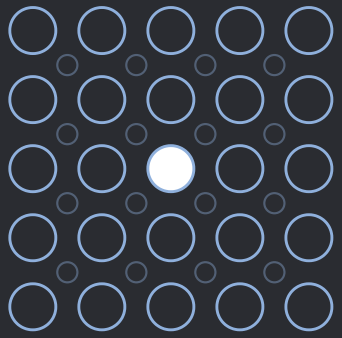
- Every lap starts with an empty context: the files are the memory
- Only works because the artifacts exist: durable spec, bounded slices, observable checks

> **Autonomy is earned by the artifacts**, not by the loop.

One slice, one owned decision

coordination_threshold = **8**

FCC bulk



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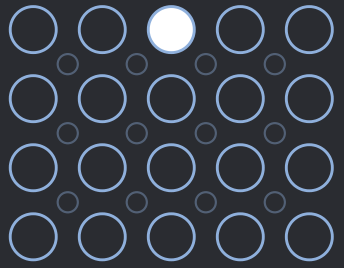
nearest neighbours

ordinary FCC(100) surface

vacuum



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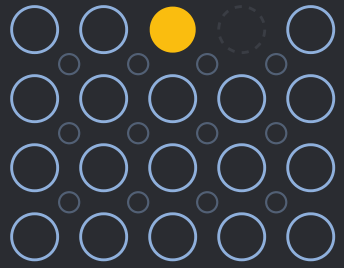
still crystal at threshold 8

beside a surface vacancy

vacuum



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below threshold → search events here

> A specification does not make the scientific decision. It **preserves** it, and turns it into explicit, testable behaviour.

Do not fix what you cannot reproduce

- First: one command that fails on the exact symptom
- Minimise it; keep it as a regression test at the right seam
- Only then: the smallest fix, and the same evidence turns green
- Re-run the original scenario

The agent must prove it sees the same bug I see.

Vocabulary: the workflow

Term	In this talk
design foundation	Durable Markdown distillation of the agreed design. A context start for future sessions.
implementation spec	The agreed end state packaged for execution. Study or method spec for simulation campaigns.
vertical slice	A narrow, complete capability, independently verifiable. A file or layer is not a slice.
red-capable reproducer	One runnable command that fails on the exact reported symptom, then passes after the fix.

A prompt earns its packaging

- A discipline I repeat, preserved as a Skill
- Exposed only to the chats where it is relevant
- Observed for drift; revised or removed when it stops helping
- Not only coding: one Skill adversarially reviews my statistical claims before I draft prose

> Not something to collect. **Something to maintain.**

Expose the decisions

`Skill: interview-design-decisions`

- The discipline: no code until the decisions are on the record
- The agent interviews me: one consequential choice at a time
- It proposes options and recommends one
- I lock in the load-bearing decisions

Surface the decisions the agent would otherwise guess.

A recorded decision is a findable mistake

HTST rates for pyKMC: the interview forced the consequential choices into the open.

- One question settled the time scale: I took the paper's convention, and the spec recorded it
- Later, a computed rate landed orders of magnitude off an accepted value
- Asking *why* did not need archaeology: the choice that went wrong was on the record

> The Skill did not catch the bug. **It made the wrong decision findable.**

The Skill, as written

No magic in the file: prose that tells the agent how to treat me.

```
Interrogate me about each decision that is not
already settled in the code base or foundation.
One question at a time. Offer real options and
say what each one changes. Wait for my decision,
then record it. Stop when I can defend every choice.
```

> It hunts the decisions **not yet on the record**.

What does this intern need today?

Study chat

Textbook read-aloud Skill: preserve the spoken text, skip figures, expand acronyms.

Hobby chat

A hobby's own Skill shelf, just as carefully maintained. Nothing a repository needs.

Coding chat

Debugging Skill: reproduce the exact failure before repairing it. What this repository needs.

> Not good Skill versus bad Skill: **context this intern should see today**, versus context it should not.

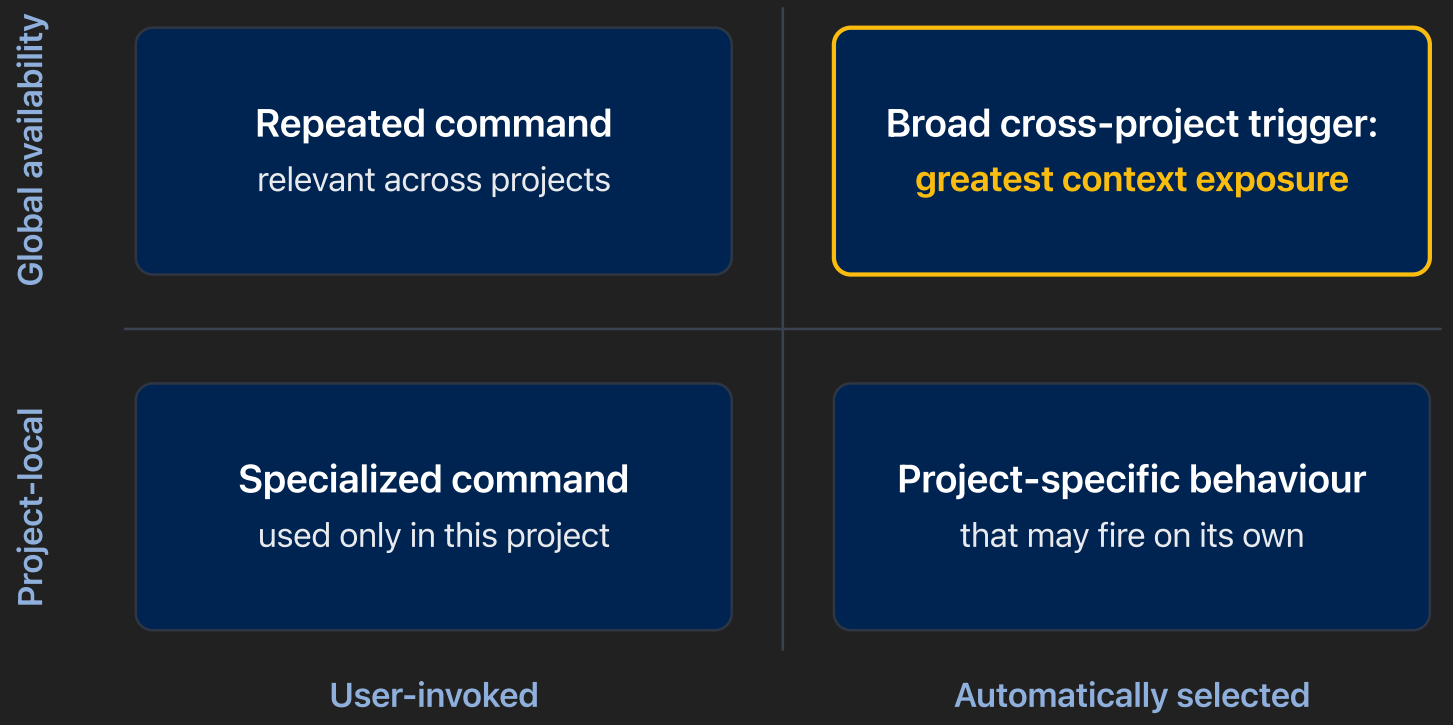
Installing a Skill expands what the agent may trust, trigger, and run

Agents are eager: give one a tool and it will look for a reason to use it.

- Listed in a CLAUDE.md or AGENTS.md file? Then every agent, and every subagent it spawns, has it on the desk
- Instructions enter the trust boundary: possibly scripts, hooks, network access

> Inspect before you adopt. Keep the active set small.

Two axes, one trigger surface



> Trust is a prerequisite at **any** scope. After that, placement is contextual relevance.

Still the wild west

- No settled practice: Skill sets, project instructions, file conventions all differ
- Different arrangements really do produce different results
- How we work now is not how we will work in a few years

My configuration is a starting point, not a standard. Keep the habit: decide what belongs in context, and change it as the tools change.

One shelf, under version control

- Every Skill I maintain lives in one repo; each project checks out only its shelf
- It helps generate new Skills tailored to a given project
- It tracks which Skills are related, so an improvement to one is disseminated to the rest

`github.com/Kerrator/skills`

> Open to the group: a jumping-off point for **a workflow of your own**.

Vocabulary: Skills

Term	In this talk
Skill	A discipline I repeat, packaged so the next chat does not rediscover it. Maintained, not collected.
trigger surface	Everything a chat can see that can cause behaviour. Every place a Skill is listed makes it larger.
invocation mode	Runs when I ask, versus selected by the model from its description.
installation scope	On one project's desk, or on every desk I own. A different axis from invocation.

Less connected to their own writing

MECH 479, across course offerings:

AI as orientation → AI as critic → AI as end-to-end author → **outer space**

- Rubric-shaped reports, produced very quickly
- Weak answers to questions about their own cited sources

> The assignment stopped guaranteeing that a polished deliverable reflected understanding. **Redesign the checkpoints, not the blame.**

Stay inside the epistemic loop

The point where I have delegated so many intermediate judgements that I can no longer audit, explain, or defend the result.

I call it outer space.

Better structure, wrong comparison

AI restructured my dipole paragraph. Genuinely better writing.

`A > B` **became** `A < B`

the shape of the error, not the sentence

> I kept the structure and repaired the science. Possible only because **I still knew what the sentence was supposed to say.**

Green tests can agree on the wrong physics

$$5 \times 10^{12}\ \mathrm{Hz} \;=\; 5\ \mathrm{ps}^{-1}$$

- The Vineyard prefactor came out in Hz; the rate layer consumed the number as ps^{-1}
- Every finite HTST rate 10^{12} too large; simulated time 10^{12} too short

The tests did not stop me. A value I already had reason to expect did.

The unit boundary, in full

$$\nu_0 = \frac{1}{2\pi\hbar} \frac{\prod_i \hbar\omega_i^{\text{min}}}{\prod_j{}' \hbar\omega_j^{\text{saddle}}}$$

- The primed saddle product omits the imaginary mode: one energy dimension survives, so ν_0 carries units
- The error multiplies every finite HTST and RPA rate; constant-style runs and the k_0 fallback were unaffected
- The fix converts $\mathrm{Hz} \to \mathrm{ps}^{-1}$ exactly once, at the backend boundary, pinned by a Ni(100) regression against a known 5 THz prefactor

Public evidence trail

github.com/hugomoison/pyKMC · verified 2026-07-13

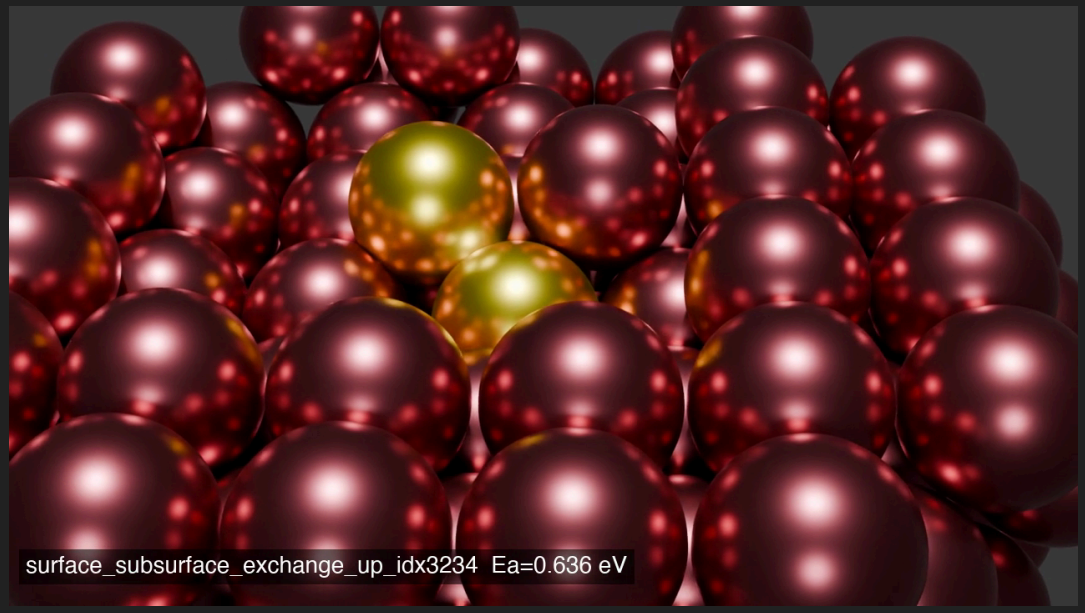
Reference	What it shows
`d705932`	Hz → ps^{-1} converted exactly once, at the backend boundary
`d6b921f`	Regression: rate, residence time, constant-versus-HTST equivalence at the same prefactor
PR 71	The historical HTST design record (a closed, unmerged draft)

> The links verify the implementations. **The disposable handoffs between them are my account.**

Vocabulary: ownership

Term	In this talk
epistemic loop	The chain of judgements you can still audit, explain, and defend.
outer space	My name for the far side: so much delegated that the result can no longer be defended.
external anchor	A value you already had reason to expect. It can catch what green tests encode wrongly.
regression vs validation	A pinned, machine-checkable expectation, versus an end-to-end confirmation run. Different strengths of evidence.

Going from idea to implementation has never been easier



blender to-do Netrunner lotusfield

Delegate labour, not scientific judgement.

- Explain the choices
- Trace the evidence
- Challenge the output
- Say what observation would prove it wrong

Thank you. Questions?